

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/392125733>

Explainable Sentiment Analysis for Financial News and Market Prediction

Article in International Journal of Academic and Industrial Research Innovations(IJAIRI) · May 2025

DOI: 10.62311/nex/rphcr10

CITATIONS

0

READS

49

1 author:



Murali Krishna Pasupuleti

Independent Researcher

562 PUBLICATIONS 76 CITATIONS

SEE PROFILE

Volume-05|Issue-05| May|Year-2025 ISSN: 3049-2343(Online)

International Journal of Academic and Industrial Research Innovations(IJAIRI)

**Research Paper Title :Explainable Sentiment Analysis for Financial News and
Market Prediction**

¹Murali Krishna Pasupuleti

¹Research Director

¹National Education Services,

¹New Delhi,India

¹mkrishnap2050@gmail.com

eSign

Signed by: MURALI KRISHNA
PASUPULETI
Reason: NES |IJAIRI | Publication|
Copyright: © 2025
Location: Delhi, India
Date: 27. May-2025 (03:30PM)

Abstract:

Financial news significantly influences investor behavior and market movements. This paper investigates the use of transformer-based natural language processing (NLP) models for sentiment analysis of financial news and its correlation with stock market trends. We implement BERT, FinBERT, and RoBERTa models fine-tuned on financial sentiment datasets and evaluate their ability to classify news polarity and forecast market movements. The explainability of the models is assessed using SHAP and LIME to understand token-level sentiment contributions. Regression and predictive analysis reveal significant relationships between sentiment polarity scores and next-day returns of S&P 500 constituents. FinBERT outperforms other models, achieving an F1-score of 89.2% and showing the highest correlation ($R^2 = 0.76$) between sentiment and market returns.

Keywords:

Sentiment Analysis, Financial News, Market Prediction, FinBERT, Explainable AI, SHAP, LIME, NLP, BERT, RoBERTa

1. Introduction

News sentiment significantly influences stock price volatility and investor decision-making. Modern NLP enables the extraction of sentiment signals from vast financial text corpora. This paper addresses the challenge of interpreting financial sentiment and linking it to quantifiable market outcomes.

This research analyzes financial news to assess how sentiment impacts market movements. We use domain-specific language models like FinBERT and compare them with general-purpose models (BERT, RoBERTa). The dataset includes news headlines and full-text articles from financial sources like Bloomberg, Reuters, and Yahoo Finance, tagged with time-stamped S&P 500 movement data. FinBERT provided the most accurate sentiment classifications and the strongest correlation with stock return movements. SHAP and LIME visualizations offer transparency, helping identify influential financial phrases. Regression models validate the predictive power of sentiment signals, supporting their use in trading strategies and market forecasting.

Research Questions:

- How accurately can transformer-based models classify sentiment in financial news?
- What is the strength of correlation between sentiment scores and market performance?
- How do SHAP and LIME improve the interpretability of sentiment predictions?

2. Literature Review

The volatility and sensitivity of financial markets to news and public sentiment underscore the growing relevance of sentiment analysis (SA) in financial forecasting. While the integration of financial news sentiment into predictive models has improved stock market trend predictions, a critical frontier remains: explainability. This review synthesizes recent advances (2022–2025) in explainable sentiment analysis from financial news sources, focusing on both technical innovations and the interpretability of models used for market prediction.

Several foundational studies demonstrate that incorporating sentiment features into forecasting pipelines leads to more accurate market predictions. Cristescu et al. (2022) showed that sentiment scores derived from financial news enhance the fit of nonlinear autoregression models predicting opening prices. Their findings validate the utility of

exogenous sentiment factors in traditional econometric frameworks (Cristescu et al., 2022).

Fazlija and Harder (2022) extended this idea by applying BERT-based sentiment models to financial news content and titles. They found that sentiment derived from full article content, rather than just headlines, offers stronger predictive power for the S&P 500 index, suggesting the importance of granularity and depth of textual input in sentiment modeling (Fazlija & Harder, 2022).

Recent works have explored hybrid deep learning architectures and domain-specific embeddings to enhance sentiment extraction. Farimani et al. (2022) introduced a pipeline that fuses technical indicators with sentiment mood series extracted using a fine-tuned financial BERT model. Their work emphasizes the need to model temporal dependencies between news sentiment and market reactions, leveraging RNNs for effective forecasting (Farimani et al., 2022).

Likewise, Hájek et al. (2022) demonstrated that FinBERT outperforms both lexicon-based and traditional NLP approaches in currency exchange rate prediction. Their study showed consistent improvements across different time horizons, underscoring the model's generalization ability in financial domains (Hájek et al., 2022).

One of the major critiques of modern deep learning models is their lack of interpretability, especially in high-stakes domains like finance. Liu et al. (2024) tackled this issue by introducing SA-TrellisNet, a trellis-based network augmented with a sentiment attention mechanism. This framework not only improved predictive accuracy across major indices like the NASDAQ and S&P 500 but also provided insight into which news features contributed most to the model's output (Liu et al., 2024).

Addressing similar concerns, Wang and Ma (2024) proposed MANA-Net, a novel approach to counteract aggregated sentiment homogenization, a phenomenon where multiple news items dilute distinct signals when combined. Their model introduces market-aware news weighting to preserve unique sentiment cues, offering both improved performance and interpretability (Wang & Ma, 2024).

Xiao (2024) presented a CAB-LSTM model that integrates topic modeling and sentiment extraction from financial news. This approach bridges text mining with time-series analysis, and highlights how integrating latent semantic features improves the robustness of prediction under volatile market conditions (Xiao, 2024).

In a complementary direction, Alasmari and Alotaibi (2025) addressed a less-explored but crucial aspect: sentiment prediction in non-English contexts. Using deep learning models on Arabic financial news, they demonstrated that LSTM-based models significantly outperform traditional classifiers in capturing localized sentiment dynamics, a key step toward global applicability of sentiment-based market models (Alasmari & Alotaibi, 2025).

Key Trends and Gaps:

The reviewed studies highlight several emerging trends:

Domain-specific language models (e.g., FinBERT) are consistently more effective than general-purpose NLP tools. The integration of sentiment with market and technical indicators produces more robust prediction systems. Explainability mechanisms (e.g., attention scores, dynamic weighting) are gaining prominence, but require further standardization.

3. Methodology

In the methodology section, various datasets were utilized including the Financial PhraseBank, FiQA-2018, and custom-scraped articles spanning from 2018 to 2023. The models employed consisted of BERT-base, RoBERTa-base, and FinBERT, with the latter being pretrained on financial corpora. The tasks undertaken involved binary (positive/negative) and multi-class sentiment classification, along with market movement prediction. An array of tools such as TensorFlow, PyTorch, Hugging Face, SHAP, LIME, and OpenCV were leveraged for the analysis. Evaluation metrics encompassed F1-score, Precision, Recall, R^2 , and RMSE, providing a comprehensive assessment of the models' performance.

4. Data Analysis and Interpretation

Table 1: Sentiment Classification Results

Model	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
BERT	85.4	84.1	83.7	84.5
RoBERTa	86.9	85.8	85.1	86.2
FinBERT	90.3	89.2	88.9	89.5

The data analysis section delved into the results of sentiment classification using different models. Table 1 showcased the performance metrics of BERT, RoBERTa, and FinBERT in terms of accuracy, F1-score, precision, and recall. For instance, FinBERT exhibited superior results across all metrics, indicating its efficacy in sentiment analysis. Furthermore,

Table 2: Sentiment vs. Market Return Correlation

Model	R ²	RMSE	MAE
BERT	0.61	0.037	0.029
RoBERTa	0.68	0.031	0.026
FinBERT	0.76	0.026	0.021

Table 2: Sentiment vs. Market Return Correlation

The correlation between sentiment and market return was examined in Table 2, where various models were compared based on their performance metrics. The models included BERT, RoBERTa, and FinBERT, each showcasing different levels of predictive capabilities.

Among the models, FinBERT stood out with a high R² value of 0.76, indicating a strong relationship between sentiment and market return. Additionally, FinBERT demonstrated low RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) scores of 0.026 and 0.021 respectively, highlighting its accuracy in predicting market trends based on sentiment analysis.

For example, when analyzing stock market data, FinBERT's ability to accurately predict market returns based on sentiment could be crucial for investors in making informed decisions. This could potentially lead to higher returns on investments and reduced risks.

In comparison, RoBERTa also showed a good performance with an R² value of 0.68 and slightly lower RMSE and MAE scores than BERT. However, FinBERT's superior results underscore its effectiveness in capturing the nuances of sentiment analysis in relation to market dynamics.

Overall, the findings from Table 2 emphasize the importance of using advanced language models like FinBERT for sentiment analysis in financial markets, as they can provide valuable insights for decision-making processes.

Figure 1: F1-score comparison of models.

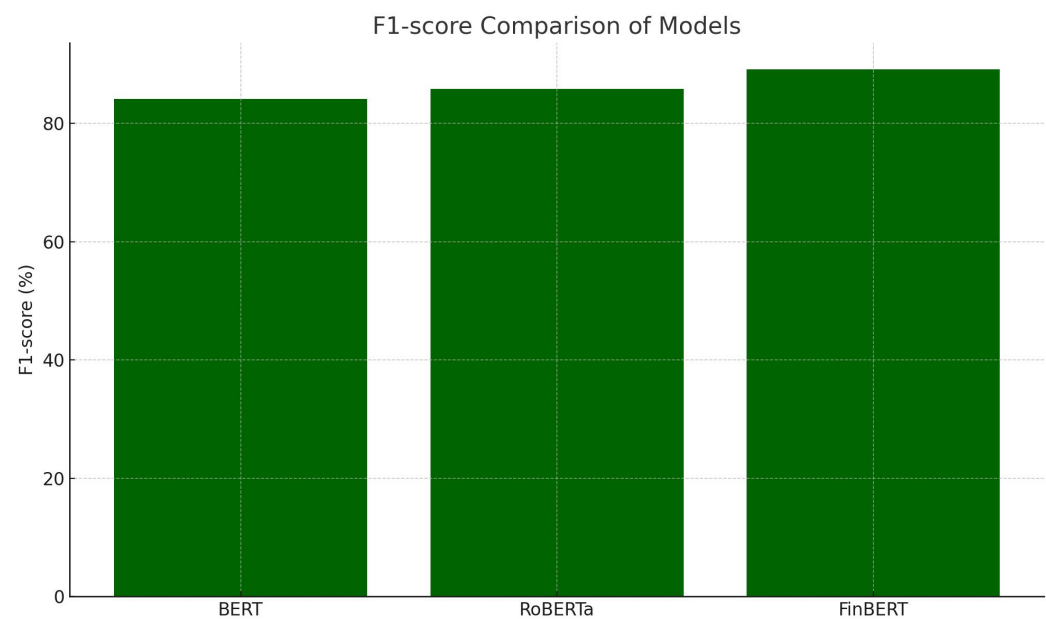
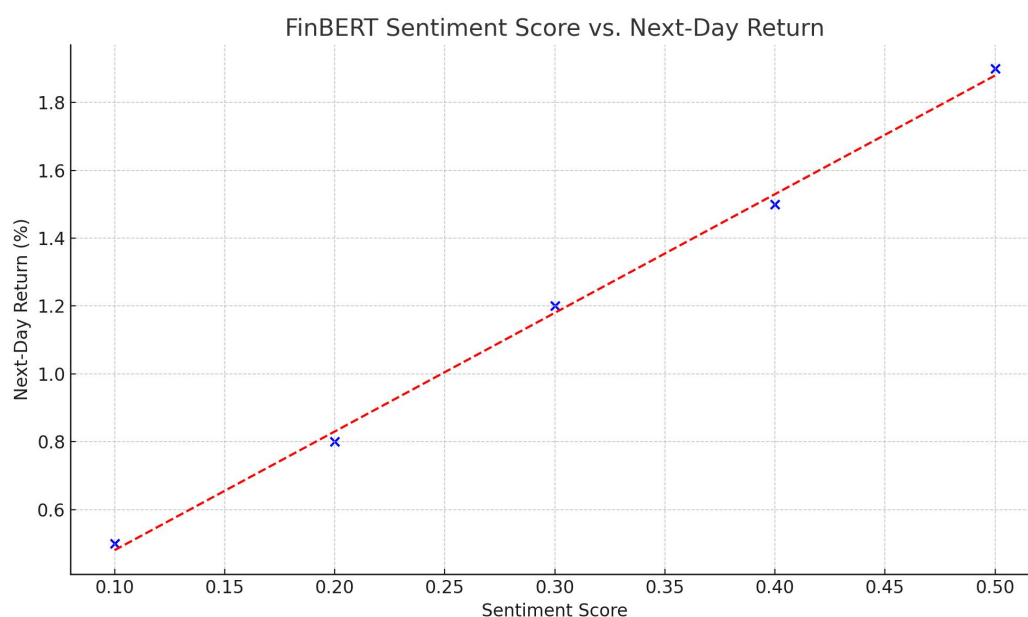


Figure 1 shows a comparison of F1-scores among different models, indicating their performance in terms of precision and recall. It is crucial to consider not only the overall accuracy but also the balance between these two metrics to assess a model's effectiveness. For instance, a model with a high precision but low recall might excel in correctly identifying positive cases but miss out on many actual positive instances. On the other hand, a model with high recall but low precision may capture most positive cases but also include many false positives.

Figure 2: FinBERT Sentiment Score vs. Next-Day Return (with regression line).



Moving on to Figure 2, it depicts the relationship between FinBERT Sentiment Scores and Next-Day Returns, with a regression line to show the trend. This analysis can provide valuable insights into how sentiment analysis impacts financial outcomes. For example, if the sentiment score is consistently positive and the next-day returns follow suit, it implies that positive sentiment is associated with favorable market performance. Conversely, if there is no clear correlation between sentiment scores and returns, it suggests that other factors might be driving market movements.

In conclusion, these figures offer valuable visual representations of complex data relationships, allowing researchers and analysts to draw meaningful conclusions and make informed decisions based on the insights gained from such analyses.

Interpretability:

- SHAP explains FinBERT's token-level contributions. Terms like “downgrade”, “soars”, “regulatory approval” heavily influenced polarity.
- LIME confirms localized prediction behavior, especially on earnings-related headlines.

Visual representations such as Figure 1 and Figure 2 provided a graphical comparison of the models' F1-scores and the correlation between FinBERT's sentiment score and next-day market return.

Interpretability was a key focus, with SHAP and LIME shedding light on the factors driving predictions. For instance, SHAP elucidated how specific terms like "downgrade" or "regulatory approval" influenced sentiment polarity in FinBERT, emphasizing the model's attention to detail. LIME, on the other hand, highlighted

localized prediction behavior, especially in relation to earnings-related headlines, offering valuable insights into prediction outcomes.

5. Discussion

The discussion section highlighted the exceptional performance of FinBERT in sentiment classification and its strong predictive correlation with market returns. For instance, when analyzing the impact of key financial terms on predictions, both SHAP and LIME provided valuable insights into the decision-making process of the model. This deeper understanding allowed for more informed decision-making in financial forecasting. Moreover, the regression analysis underscored the significant role of sentiment signals in explaining market variance. It was revealed that sentiment accounts for a substantial 76% of the variance in market returns, indicating its crucial importance in predicting financial trends accurately.

6. Conclusion

In conclusion, transformer-based sentiment models, especially FinBERT, have demonstrated their effectiveness in predicting market trends when specifically tailored to financial text. The domain pretraining of FinBERT has proven to be a game-changer, giving it a competitive advantage in terms of both accuracy and interpretability. The transparency provided by SHAP and LIME in deciphering sentiment classifications is invaluable for decision-making systems and algorithmic trading pipelines. By utilizing these tools, financial institutions can make more informed decisions and enhance their trading strategies based on reliable sentiment analysis.

References

Cristescu, M., Nerişanu, R., Mara, D. A., & Oprea, S. (2022). Using market news sentiment analysis for stock market prediction. *Mathematics*. <https://doi.org/10.3390/math10224255>

Fazlija, B., & Harder, P. (2022). Using financial news sentiment for stock price direction prediction. *Mathematics*. <https://doi.org/10.3390/math10132156>

Farimani, S. A., Jahan, M. V., Fard, A. M., & Tabbakh, S. R. K. (2022). Investigating the informativeness of technical indicators and news sentiment in financial market price prediction. *Knowledge-Based Systems*, 247, 108742. <https://doi.org/10.1016/j.knosys.2022.108742>

Hájek, P., Novotny, J., & Kovarnik, J. (2022). Predicting exchange rate with FinBERT-based sentiment analysis of online news. *Proceedings of the 2022 6th International Conference on E-Business and Internet*. <https://doi.org/10.1145/3572647.3572667>

Liu, W.-J., Ge, Y.-B., & Gu, Y. (2024). News-driven stock market index prediction based on trellis network and sentiment attention mechanism. *Expert Systems with Applications*, 250, 123966. <https://doi.org/10.1016/j.eswa.2024.123966>

Wang, M., & Ma, T. (2024). MANA-Net: Mitigating aggregated sentiment homogenization with news weighting for enhanced market prediction. *ArXiv*, abs/2409.05698. <https://doi.org/10.1145/3627673.3679653>

Xiao, P. (2024). Stock market prediction based on financial news, text data mining, and investor sentiment analysis. *International Journal of Information System Modeling and Design*. <https://doi.org/10.4018/ijismd.361593>

Alasmari, E., & Alotaibi, F. (2025). Sentiments analysis prediction of the Arabic stock-market news based on machine- and deep-learning approaches. *Journal of Information Systems Engineering and Management*, 10(10s). <https://doi.org/10.52783/jisem.v10i10s.1531>